AI-Assisted Ensemble Techniques

1.1 Project Setup and Business Problem Understanding

- Set up a minimal project directory for a click prediction case study covering EDA, preprocessing, feature engineering, and modelling.

Note: *Place the provided dataset in the data/raw/ folder before moving to the next step. Refer to the Business Context document and review the business context and objective before continuing.*

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Business Context

ShopSpark Retail is a mid-size US e-commerce company that sells across eight product categories. It serves customers through a website and a mobile app, with mobile making up around 77 percent of traffic. The company has 8,000 active customers and a catalog of 704 products. Customers range from new sign-ups to loyal repeat buyers, spread across major US metros (New York, Los Angeles, Chicago, Houston, Dallas, Phoenix, Philadelphia, San Antonio, San Diego) and a long tail of smaller cities.

On every page a shopper visits, ShopSpark fills several recommendation slots with products. Today, these slots are filled using simple rules: best-sellers in the category, most-viewed in the last week, or items that are on discount. The product team believes the company is leaving money on the table. The same homepage carousel is shown to a 22-year-old fashion shopper in New York and to a 50-year-old home goods buyer in Phoenix. The team wants a data-driven way to predict which product is most likely to be clicked by which user in which context, so that, in a later phase, the ranking on each slot can be driven by that prediction.

Objective

To build a model that predicts the probability of a click for any product shown to any shopper in any context, so that recommendation slots across the website and app can be ranked by predicted click probability instead of today's static rules.

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1.2 Data Understanding and Joining Tables

- Create a new notebook 01_data_understanding.ipynb in the notebooks folder.
- @01_data_understanding.ipynb Load the three CSVs from data/raw/, join them into one dataset on user_id and product_id, save the joined view to data/interim/, and summarise shape, column types, missing values, and click rate.

1.3 Exploratory Data Analysis

- Create a new notebook 02_eda.ipynb in the notebooks folder.
- @02_eda.ipynb Load the joined dataset and do univariate analysis on the target column clicked and on price_usd.
- @02_eda.ipynb Add bivariate analysis. Click rate by platform and click rate by page_type.
- @02_eda.ipynb Add two multivariate plots. Heatmap of click rate by category and platform, and heatmap of click rate by hour of day and day of week.

1.4 Metric Selection and Data Splits

- Split the dataset into train (70 percent), validation (15 percent), and test (15 percent). Use a fixed random seed. Stratify on the target. Print shape and click rate for each split.

1.5 Data Preprocessing

- Create a new notebook 03_preprocessing.ipynb in the notebooks folder.
- @03_preprocessing.ipynb Load the train, validation, and test splits. Propose a preprocessing plan covering missing values, outliers in price_usd, and encoding for categorical columns. The plan should fit each transform on the train split only and apply the same parameters to validation and test. Do not run yet.
- @03_preprocessing.ipynb The plan looks good. Apply it: fit the p99 cap and the encoders on the train split, then apply the fitted transforms to validation and test. Save the cleaned train/val/test splits to data/processed/ and confirm.

1.6 Feature Engineering

* Create a new notebook 04_feature_engineering.ipynb in the notebooks folder.
* @04_feature_engineering.ipynb Load the cleaned train, validation, and test splits. Propose 6 derived features that would help predict clicks. For each, explain how it is computed, what it captures, and how to fit on the train split then apply the same transformation to validation and test. Do not run yet.
* @04_feature_engineering.ipynb The plan looks good. Build all six features by fitting any required parameters on the train split and applying the same transforms to validation and test. Save all three final splits to data/processed/ and summarise.

1.7 Model Building

* Create a new notebook 05_model_building.ipynb in the notebooks folder.
* @05_model_building.ipynb Train a Random Forest classifier with class_weight set to balanced. Report Train and Validation AUC-ROC.
* @05_model_building.ipynb Tune the Random Forest with 3 best parameters of 4 combinations. Report Train and Validation AUC for each combination and pick the best.
* @05_model_building.ipynb Train a Gradient Boosting classifier with defaults. Report Train and Validation AUC-ROC.
* @05_model_building.ipynb Tune the Gradient Boosting with 3 best parameters of 4 combinations. Report Train and Validation AUC for each combination and pick the best.
* @05_model_building.ipynb Train an XGBoost classifier with scale_pos_weight set for class imbalance. Report Train and Validation AUC-ROC.
* @05_model_building.ipynb Tune the XGBoost with 3 best parameters of 4 combinations. Report Train and Validation AUC for each combination and pick the best.

1.8 Model Comparison and Selection

* @05_model_building.ipynb Build a comparison table for all three models. Pick the best based on validation AUC and the train-validation gap.

1.9 Test Performance

* Create a new notebook 06_test_performance.ipynb in the notebooks folder.

- @06_test_performance.ipynb Load the best model. Evaluate it on the test set. Report Test AUC-ROC and the classification report at threshold 0.5.
- @06_test_performance.ipynb Plot the top 15 feature importances from the best model.

1.10 Top 5 Product Rankings and Business Report

- Create a new notebook 07_rankings_and_report.ipynb in the notebooks folder.
- @07_rankings_and_report.ipynb Pick one representative user. Score every product in the catalog for that user in a fixed context (mobile_app, homepage, Tuesday 7 PM, slot 1). Show the top 5 products with name, brand, category, price, and predicted click probability.
- Create an HTML report at reports/shopspark_recommendation_report.html that includes the user profile, the top 5 ranking, model performance summary, what drives the prediction, three concrete business actions with supporting evidence and expected impact, and a short validation summary confirming the objective is met.